Ahmadreza Farvardin

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EDUCATION

University of Tehran Tehran, Iran
B.Sc. in Mathematics & Computer Science Expected: June 2027
Major GPA: 18.05/20 (Top 5%) | Minor GPA: 17.23/20 | Ranked 2nd out of 40 in Mathematics
Relevant Coursework: Machine Learning, Deep Generative Models, Nonlinear Optimization, Linear Algebra, Statistical Modeling, Mathematical Analysis

RESEARCH INTERESTS

Multimodal Large Language Models, Embodied AI, Generative Models, Computer Vision

TECHNICAL SKILLS

Programming: Python (Advanced), C/C++, MATLAB, Bash
ML/DL Frameworks: PyTorch, Scikit-learn, NumPy, Pandas, OpenCV, Matplotlib
Deep Learning: Transformers, CNNs (ResNet, U-Net, YOLO, Faster R-CNN), Attention Mechanisms, LLMs, Vision-Language Models, Inference Optimization
Specializations: Computer Vision (Segmentation, Detection, Optical Flow), NLP (Machine Translation), Generative Models
Mathematics: Optimization Algorithms, Linear Algebra, Statistical Modeling, Numerical Methods
LLM Methods & Evaluation: LLM-driven program search, Evolutionary algorithms (MAP-Elites), Prompt optimization, Evaluation design, Held-out validation, McNemar testing, Specification gaming analysis
Tools: Git, Linux, JupyterLab, VS Code, LaTeX

RESEARCH & PROJECTS

AlphaEvolve-VLM: Evolving Vision-Language Inference 2026
Princeton NLP Lab – Technical Assessment

- Built an AlphaEvolve-style program-evolution system from scratch. It has an island model with MAP-Elites, diff-based LLM mutation using Gemini Flash, staged GPU evaluation, AST anti-cheating checks, and crash-safe resume. It evolves the inference code of Qwen3-VL-2B for chart question answering on CharXiv.
- Hand-optimized the Instruct model from 25.9% to 56.0% holdout accuracy while making it 6× faster, through one-change-at-a-time experiments: answer extraction, answer-only prompting, and abstention checks.
- Evolution added +6.0 points over hand-tuning on an untouched 800-query holdout (McNemar p = 0.0003). It also made the reasoning model 33% faster and 2.2 points more accurate (67.0%, the best result), costing only $0.23 in LLM calls.
- Found specification gaming and a validation error through strict evaluation. An answer-level audit showed that evolution's whole gain came from exploiting the benchmark's "Not Applicable" base rate (+91 / −43 answers). The holdout split also overturned a two-model routing design that had been validated against the wrong baseline.
- Technologies: Python, PyTorch, LLMs, Evolutionary Search, Evaluation Rigor.

River Velocity Estimation Using Deep Learning & Classical PIV 2024
Computer Vision Research Project

- Developed hybrid pipeline combining RAFT optical flow (Princeton) and OpenPIV cross-correlation for automated river surface velocity measurement from video footage
- Implemented end-to-end system: video preprocessing, RAFT-based flow estimation for unseeded videos, classical PIV for seeded flows, and cross-validation calibration across camera setups
- Achieved $R^2 = 0.77$, $MAE = 0.36 m/s$, $RMSE = 0.58 m/s$ on 22 real-world river videos with varying conditions
- Technologies: PyTorch, RAFT, OpenPIV, OpenCV, NumPy

Deep Learning Architecture Implementations from Scratch 2024–2025
Advanced ML Engineering Projects

- **Transformer for Machine Translation :** Implemented complete Transformer (Vaswani et al., 2017) from primitives for German→English translation on Multi30k dataset, including multi-head attention, positional encoding, masking mechanisms, and Noam scheduler
- **U-Net with Attention Gates :** Built U-Net for binary semantic segmentation on Oxford-IIIT Pets dataset; achieved 0.8375 IoU and 0.9116 Dice score
- **YOLOv3 Object Detection :** Developed YOLOv3 from scratch on COCO dataset without pretrained weights, implementing anchor boxes, multi-scale predictions, and custom loss functions
- **ResNet-18 Image Classification :** Implemented ResNet-18 architecture from basic components for CIFAR-10
- Technologies: PyTorch, NumPy, OpenCV

Persian Handwritten Mathematical Expression Recognition 2024

Machine Learning Course Project — 1st Place, Quera Competition

- Designed end-to-end pipeline: YOLOv8/Faster R-CNN for symbol detection, K-Means/DBSCAN for clustering, CRNN + CTC loss for recognition
- Achieved 92.3% mAP on detection; implemented semi-supervised learning for data efficiency
- Technologies: PyTorch, OpenCV, NumPy, Pandas

Machine Learning & Optimization Algorithms 2024

Course Projects (ML | Optimization)

- Implemented core ML algorithms from scratch: Naive Bayes, Ridge/Logistic Regression, AdaBoost, PCA, GMM
- Developed optimization algorithms: Gradient Descent, Newton's Method, Regularized Least Squares with applications to SVMs and polynomial regression
- Technologies: Python, NumPy, Pandas, Scikit-learn, cvxpy, Matplotlib

TEACHING EXPERIENCE

Teaching Assistant University of Tehran

School of Electrical & Computer Engineering and School of Mathematics, Statistics & CS

- **Nonlinear Optimization** — Supervising Professor: Prof. Mohammad Saeed Sarafraz Fall 2024
- **Linear Algebra** — Supervising Professor: Prof. Mohammad Javad Yazdanpanah Spring 2024
- **Mathematical Analysis** — Supervising Professor: Prof. Fatemah Ayatollah Zadeh Shirazi Spring 2024
- **Calculus I & II** — Supervising Professor: Prof. Shaghayegh Mokarami Fall 2023 – Spring 2024
- Conducted weekly problem-solving recitations, designed assignment solutions, and held regular office hours for undergraduate engineering and mathematics students.

HONORS & AWARDS

- **1st Place** — Quera Competition for Machine Learning Final Project (Mathematical Expression Recognition), 2024
- **Ranked 2nd out of 40** — Mathematics Major, University of Tehran

ADDITIONAL INFORMATION

Languages: Persian (Native), English (Professional Working Proficiency)